Ethan Nguyen

EthanNguyen112@gmail.com | +1 (714) 913-5777
LinkedIn: Ethan-Nguyen112 | GitHub: EthanNguyen112 | Website: EthanNguyen.xyz

Skills

Embedded & Hardware: C, C++, Verilog, Vivado, Arduino IDE, FPGA, UART, SPI, PWM, ADC

Software & Web: Python, TypeScript, Node.js, React, Next.js, Tailwind, NestJS, Prisma, PostgreSQL, REST APIs, JWT, RBAC

Tools & Platforms: Git, Jira, Bitbucket, Confluence, Bash, GCP, Swagger, VBScript, Visio

Experience

Caretaker Provider, In-Home Support Services – Fountain Valley, CA *August 2023 – Present*

- Diagnosed and resolved hardware and software issues across multiple devices, restoring digital accessibility for clients with limited technical background.
- Configured collaboration tools (Viber, Google Chat) to streamline daily care coordination.

Full-Stack Software Engineer (Volunteer Contributor), MuseumSoft – Remote *July 2025 – December 2025*

- Built REST APIs (NestJS, Prisma, PostgresQL, Swagger) for ticketing, CRM, memberships, and event scheduling.
- Developed responsive frontend pages and reusable components with Next.js, shadcn/ui, and Tailwind CSS.
- Implemented JWT authentication with refresh token rotation and role-based access control (RBAC).
- Built a signed-URL upload pipeline sending files directly to Google Cloud Storage, reducing backend load.

Software Engineer Intern, L3Harris Power Paragon Inc – Anaheim, CA *June 2019 – January 2021*

- Tested and debugged hardware/software interfaces with provided test cases for nuclear power control systems.
- Built a GUI-based VBScript tool to standardize documentation, reducing manual effort by ~30%.
- Automated daily status reports for 100+ Jira tickets, reducing manual reporting and improving project visibility.

Projects

FPGA Oscilloscope | Verilog

- Built a dual-channel oscilloscope from RTL logic using a 12-bit 1 MSPS XADC with real-time waveform rendering.
- Designed an analog front end scaling 0–40V inputs to 0–1V ADC range with attenuation and protection circuitry.
- Implemented ADC control, RAM buffering, edge triggering, and a custom VGA controller with adjustable voltage/div, time/div, and trigger controls.

Embedded Weather Station | C

- Built an IoT weather client on TM4C123 using CC3100 WiFi to fetch real-time data via REST API over HTTP.
- Developed a UART query interface (city, ID, coordinates, ZIP) at 115200 baud with JSON parsing.
- Designed frame-based dynamic weather animations on ST7735 LCD via SPI for smooth real-time display updates.

Bluetooth Remote Controlled Car | C

- Integrated HC-05 Bluetooth module via UART with AT command configuration for reliable wireless motor control.
- Developed a PWM-based differential drive for directional steering and variable speed control.

Education

California State University, Long Beach – Bachelor of Science in Computer Engineering

Relevant Coursework: Embedded Systems, Operating Systems, SOC Design, Digital Signal Processing, Digital Design & Tech Verification, Computer Logic Design, Microprocessors & Controllers, Computer Architecture